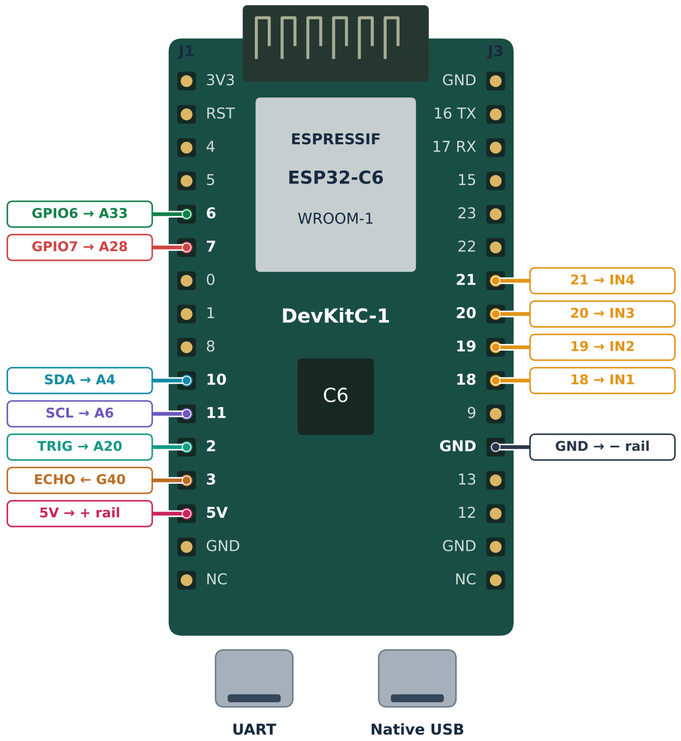
BatMetal: sonar, LCD, and LEDs

Long wires are shown as labels. Each label names the GPIO pin or breadboard hole to connect.

ESP32-C6-DevKitC-1-N8

Top view: components facing up, USB ports at the bottom.



HOW TO READ THE LABELS

Each label replaces a long wire.

Example: “GPIO6 → A33” on the board
connects to breadboard hole A33, shown in green.

Right − rail: common ground. Right + rail carries board 5 V.

LED PLACEMENT

Red: anode B39 (+), cathode B38 (−).

Green: anode D47 (+), cathode D48 (−).

220 Ω per LED; three 1 kΩ resistors for ECHO.

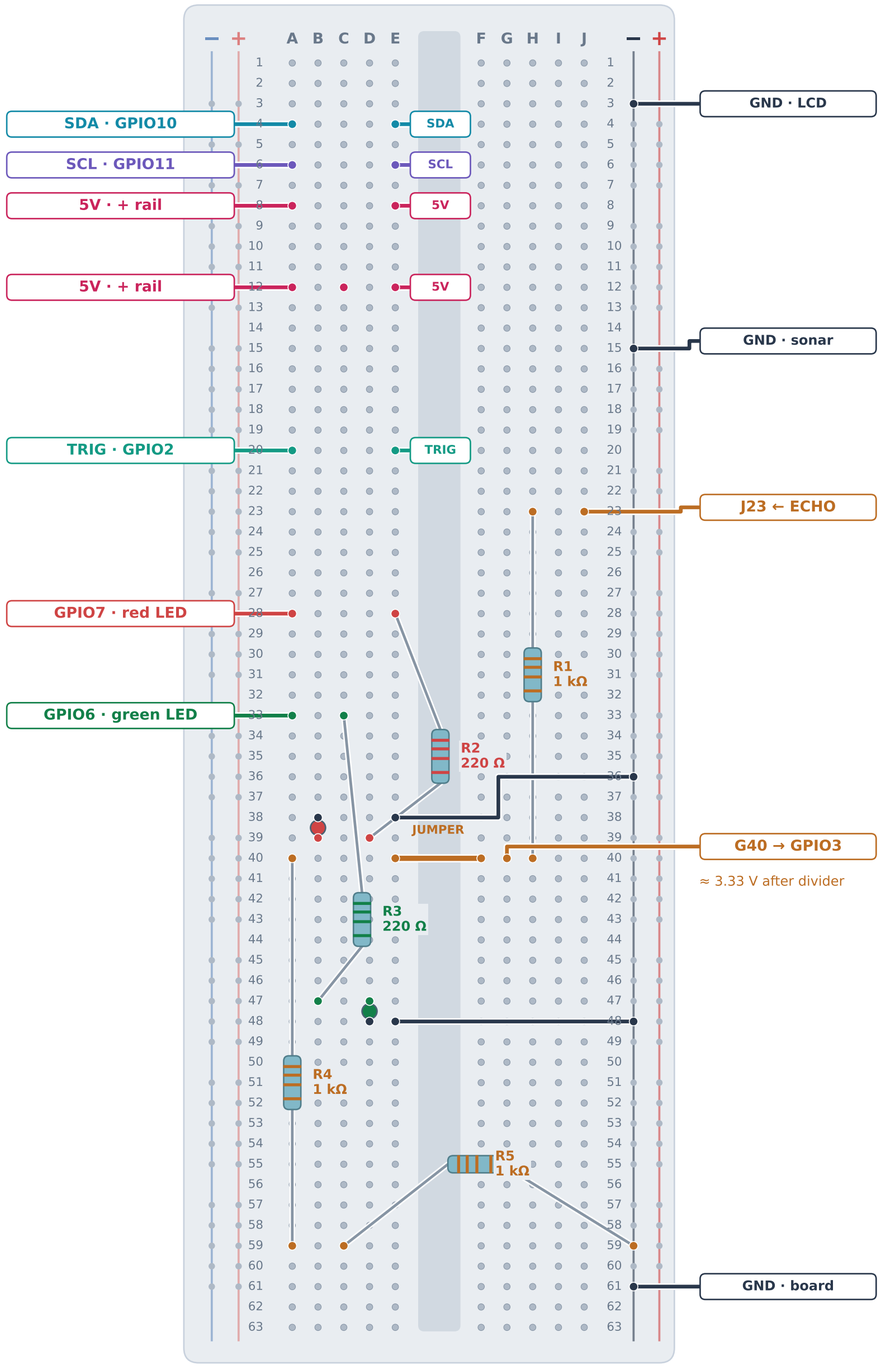
Jumper E40 ↔ F40 joins the divider node.

Dots mark connections. Crossings without dots are not connected.

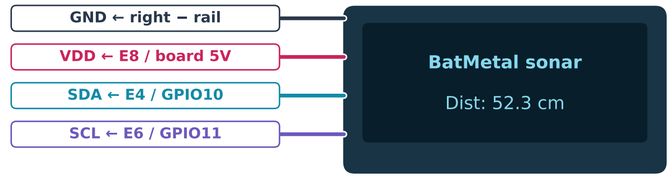
Only component leads inserted into holes make contact.

BREADBOARD: resistor positions kept unchanged

Lead positions estimated from photos; verify before wiring.



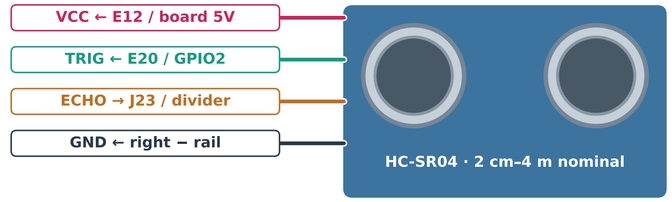
LCD · 1602 I²C, 2 lines



Keep I²C pull-ups at or below 3.3 V.

Follow printed pin names, not the illustrated order.

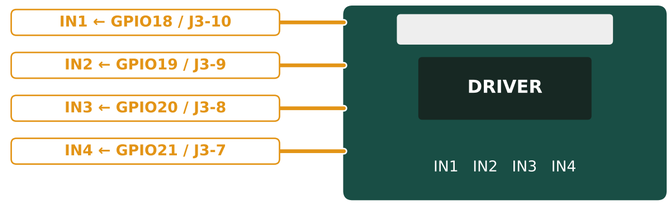
SONAR · wiring for a standard HC-SR04



ECHO → R1 → GPIO3 → R4 → R5 → GND

Nominal range; target shape and angle affect detection.

MOTOR DRIVER · 5 V 28BYJ-48



Driver previously smelled burnt; keep disconnected.

Do not power until the driver has been inspected.

Motor modes are experimental; no successful test yet.

Never connect motor coils directly to ESP32 GPIOs.

USB POWER · sonar and LCD

Board 5V → right + rail → E12 sonar and E8 LCD VDD.

Board GND → − rail → sonar, LCD, and LED grounds.

LCD is powered at 5 V; text is clearly visible.

Check SDA/SCL pull-ups stay at or below 3.3 V.

Check resistor leads: R1 H23–H40; R2 E28–D39; R3 C33–B47; R4 A40–A59; R5 C59–right − rail.

A–E and F–J are separate groups on each row. Split rails: bridge +/+ and −/− as needed. Disconnect power before wiring.

Firmware: LEDs, sonar, and LCD1602. Display text and sonar readings have been confirmed.

Board powered by USB. Sonar and LCD use board 5 V. Keep I²C signal levels safe for the ESP32-C6.